

Dropout2d#

[https://pytorch.org/docs/stable/generated/torch.nn.modules.dropout.Dropout2d](https://pytorch.org/docs/stable/generated/torch.nn.modules.dropout.Dropout2d.html)

Description:

Randomly zero out entire channels. A channel is a 2D feature map, e.g., the jjj -th channel of the iii -th sample in the batched input is a 2D tensor $\text{input}[i,j]$. Each channel will be zeroed out independently on every forward call with probability p using samples from a Bernoulli distribution. Usually the input comes from `nn.Conv2d` modules. As described in the paper *Efficient Object Localization Using Convolutional Networks*, if adjacent pixels within feature maps are strongly correlated (as is normally the case in early convolution layers) then i.i.d. dropout will not regularize the activations and will otherwise just result in an effective learning rate decrease.

Function Signature

```
class torch.nn.modules.dropout.Dropout2d(p=0.5,  
inplace=False)[source]#
```

Parameters

Shape:

```
forward(input)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.Dropout2d(p=0.2)  
>>> input = torch.randn(20, 16, 32, 32)  
>>> output = m(input)
```

Notes & Warnings

WARNING

Due to historical reasons, this class will perform 1D channel-wise dropout for 3D inputs (as done by `nn.Dropout1d`). Thus, it currently does NOT support inputs without a batch dimension of shape (C,H,W) (C, H, W) (C,H,W) . This behavior will change in a future release to interpret 3D inputs as no-batch-dim inputs. To maintain the old behavior, switch to `nn.Dropout1d`.

Detailed Documentation

Documentation

Randomly zero out entire channels.

A channel is a 2D feature map, e.g., the j -th channel of the i -th sample in the batched input is a 2D tensor $\text{input}[i,j]$.

Each channel will be zeroed out independently on every forward call with probability p using samples from a Bernoulli distribution.

Usually the input comes from `nn.Conv2d` modules.

As described in the paper *Efficient Object Localization Using Convolutional Networks*, if adjacent pixels within feature maps are strongly correlated (as is normally the case in early convolution layers) then i.i.d. dropout will not regularize the activations and will otherwise just result in an effective learning rate decrease.

In this case, `nn.Dropout2d()` will help promote independence between feature maps and should be used instead.

`p` (float, optional) – probability of an element to be zero-ed.

`inplace` (bool, optional) – If set to `True`, will do this operation in-place

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Input: `(N,C,H,W)` `(N, C, H, W)` `(N,C,H,W)` or `(N,C,L)` `(N, C, L)` `(N,C,L)`.

Output: `(N,C,H,W)` `(N, C, H, W)` `(N,C,H,W)` or `(N,C,L)` `(N, C, L)` `(N,C,L)` (same shape as input).

Runs the forward pass.